Sahaj Bhadja

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Summary

Software Engineer with an MS in CS and 2+ years writing **automated tests** and debugging production systems in **Python**, building on **LLM inference APIs**, and running **model evaluations** and benchmarks, comfortable on Linux, Git, and Docker.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Advanced NLP, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant async **LLM job orchestration** with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Added validation and reconciliation checks to integration pipelines across PostgreSQL, SQL Server, and cloud storage, sustaining 99%+ **production reliability** over 5K+ monthly jobs for 500+ users with CI/CD-backed releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built production integrations against GPT-4o, Claude, and Qwen-VL **inference APIs** with prompt chains for OCR, rubric generation, and grading, serving 10K+ daily REST requests for 150+ instructors.
- Debugged OCR reliability and timeout **failures** by adding fallback routing across providers, logged per-call token counts and cost, and sustained 90%+ grading consistency across 1K+ file batches under peak load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Trained a GPT-based Decision Transformer in PyTorch on features from 100K+ autonomous-driving trajectory logs through reproducible Hydra and W&B experiment pipelines, improving long-horizon planning 18% in closed-loop **simulation evaluation**.
- **Benchmarked** a spatial-temporal attention encoder against linear, Wayformer-style, and MotionLM-style baselines in closed-loop simulation rollouts, improving planning stability 15% and producing fewer off-road trajectories in qualitative closed-loop evaluation.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ automated unit and integration **test** suites with **PyTest** and Mockito covering end-to-end sync flows, run nightly by Jenkins on Linux, Windows, and macOS VMs with CSV reports.
- Automated nightly pipelines that pulled commits, built and installed executables, and emailed coverage summaries and **failure logs**, then troubleshot large-scale API integrations to improve response times 30% for low-bandwidth users.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Automated unit, integration, and end-to-end **tests** with Jest, Mocha, and Selenium in Jenkins CI/CD before Docker packaging and staged deployment, reaching 95%+ build validation across release runs.
- Diagnosed production issues with Sentry error tracking, Azure Monitor performance metrics, and Kibana logs across eight REST and WebSocket **microservices** synchronizing 2M+ records across multiple locations in real time.

Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash
Testing: PyTest, Mockito, Jest, Mocha, Selenium, Unit Testing, Integration Testing, End-to-End Testing, Postman
APIs & LLMs: REST APIs, HTTP, JSON, WebSockets, LLM Inference APIs, Claude, GPT-4o, Prompt Chaining
ML & Evaluation: PyTorch, Transformers, Model Evaluation, Benchmarking, Hydra, Weights & Biases
Infrastructure: Linux, Git, Docker, Kubernetes, AWS, Azure, Jenkins, GitHub Actions, CI/CD
Debugging & Observability: Sentry, Kibana, Azure Monitor, Log Analysis, Root Cause Analysis

Projects

LLM Inference API Test and Load Harness | *Python, pytest, requests, locust, Docker*

- Designed a **pytest** suite for an OpenAI-compatible inference API checking schema compatibility, streaming SSE chunk order, and error codes, plus locust **load tests** reporting time-to-first-token and tokens per second.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2, Flask*

- Evaluated BLIP, GIT, and ViT-GPT-2 multimodal models against a custom model on 10K images and 100K captions, raising caption **accuracy** from 81% to 96% behind real-time inference APIs.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*

- Implemented six scheduling algorithms in **C++** and tested correctness across preemptive and non-preemptive edge cases over 1K+ process combinations, cutting debugging time 20% with integrated Gantt chart visualization of execution.